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LiteTest Runner Reference

Runner Version: 1.0.0

Test Version: 1.0.0

This document provides a reference to the LiteTest Runner API. It includes types, structs, unions, enums, macros, and functions. Additionally, it provides a reference to LiteTest framework concepts.

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Preface

This document is intended for LiteTest user and contributors who need a reference for the LiteTest Runner framework and API.

For a list of other LiteTest documents and the repository layout, see the LiteTest Documentation Guide (`LiteTest_Documentation_Guide.md`).

For a glossary of terms, see the LiteTest Glossary Reference (`LiteTest_Glossary_Reference.md`).

Document Version History

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- The **Document** column records the document's version with the format **M.u** (Major, update).
- The **Runner** column records the LiteTest Runner API version current at the time this document version was published and is defined by its `LT_RUNNER_VERSION` macro.
- The **Test** column records the LiteTest Test API version current at the time this document version was published and is defined by its `LT_TEST_VERSION` macro.
- Both the Runner API and Test API use the version format "**M.m.p**" (Major, minor, patch).
- **M** is the same for the Document, Runner, and Test versions.

The document's update version tracks updates to this document and does not correspond to a minor or patch version. **u** increments whenever this document is updated without a change to **M**, and it resets to 0 when **M** is incremented.

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1. Introduction

TODO: add introduction, common rules.

2. Types

2.1. lt_result_t

Type for result counters.

Note: A test group function or test with a fault sets its result with the fault count set to `SIZE_MAX` to indicate a function-level fault.

See also: - `lt_currentresult(void)` in Customization Helper Functions. - `LT_GROUP(...)` notes on fault capture semantics and how function-level faults are represented.

```
typedef struct
{
    size_t pass;
    size_t fail;
    size_t fault;
    size_t injected_fail;
    size_t injected_fault;
} lt_result_t;
```

2.2 lt_state_t

Type for maintaining the state of the orchestrator (main) or a test function.

```
typedef struct
{
    size_t id;
    char *funcname;
    size_t current_level;
    size_t groupid;
    char *groupname;
    int isolation; // 0 none, 1 thread, 2 process.
    size_t category_id;
    size_t num_results_merged;
    size_t pass;
    size_t fail;
    size_t fault;
    size_t injected_fail;
    size_t injected_fault;
    size_t total_pass;
    size_t total_fail;
    size_t total_fault;
    size_t total_injected_fail;
    size_t total_injected_fault;
    lt_state_t parent;
    lt_state_t prev;
    lt_state_t next;
} lt_state_t;
```

3. Enums

3.1. `lt_exit_code_t`

Exit codes are grouped into classes: - `LT_EXIT[_*]` normal exit codes. - `LT_TEST[_]*` test outcome codes. - `LT_FATAL_USAGE[_*]` fatal usage codes (invalid arguments, invalid caller state, API misuse). - `LT_FATAL_INTERNAL[_*]` fatal internal codes (assert/invariant/internal failures). - `LT_FATAL_SYSTEM[_*]` fatal system codes (OS/runtime/resource failures).

Fatal codes indicate that the runner could not complete normal or test execution.

`LT_FATAL_USAGE`, `LT_FATAL_INTERNAL`, and `LT_FATAL_SYSTEM` are general codes for when a more specific code does not apply for the class.

```
typedef enum
{
    // 0-99: Exit codes.
    LT_EXIT_OK = 0,

    // 0-99: Test outcome codes.
    LT_TEST_PASS = 0,
    LT_TEST_FAIL = 1,
    LT_TEST_FAULT = 2,
    LT_TEST_FAIL_FAULT = 3,

    // 100-199: Fatal usage codes.
    LT_FATAL_USAGE = 100,
    LT_FATAL_USAGE_INVALID_ARG = 101,
    LT_FATAL_USAGE_INVALID_STATE = 102,

    // 200-299: Fatal internal codes.
    LT_FATAL_INTERNAL = 200,
    LT_FATAL_INTERNAL_ASSERT = 201,
    LT_FATAL_INTERNAL_INVARIANT = 202,

    // 300-399: Fatal system codes.
    LT_FATAL_SYSTEM = 300,
    LT_FATAL_SYSTEM_OPEN = 301,
    LT_FATAL_SYSTEM_READ = 302,
    LT_FATAL_SYSTEM_WRITE = 303,
    LT_FATAL_SYSTEM_FORK = 304,
    LT_FATAL_SYSTEM_THREAD = 305
} lt_exit_code_t;
```

3.1. `lt_return_code_t`

Return codes are for functions that return an integer (e.g., `int`) and are grouped into classes: - Success - Compare - Boolean - `LT_INVALID[_*]` invalid usage codes (invalid arguments). - `LT_SYSTEM[_*]` failed system codes (OS/runtime/resource failures).

LT_INVALID and LT_SYSTEM) are general codes for when a more specific code does not apply for the class.

```
typedef enum
{
    // Success
    LT_OK = 0,

    // Compare
    LT_LESS = -1,
    LT_EQUAL = 0,
    LT_GREATER = 1,

    // Boolean
    LT_FALSE = 0,
    LT_TRUE = 1,

    // -100 to -199: Invalid Usage.
    LT_INVALID = -100,
    LT_INVALID_ARG = -101,
    LT_INVALID_ARG_VERSION = -102,
    LT_INVALID_ARG_TOO_LONG = -103,

    // -300 to -399: Failed system call.
    LT_SYSTEM = -300,
    LT_SYSTEM_OPEN = -301,
    LT_SYSTEM_READ = -302,
    LT_SYSTEM_WRITE = -303,
    LT_SYSTEM_FORK = -304,
    LT_SYSTEM_THREAD = -305
} lt_return_code_t;
```

4. Macros for Limits

Maximum values for path length, filename length, and guard levels.

```
#define LT_MAX_PATH_LEN ((size_t)4096) #define LT_MAX_FILENAME_LEN  
((size_t)255) #define LT_MAX_LEVEL ((size_t)32)
```

Note: The limit of 32 levels is unlikely to be exceeded if there are 2 or more `LT_GROUP` or `LT_TEST` macros at each level. It is expected that a level will generally have 2 or more per level.

5. Macros for the Orchestrator Function

Macros used to define or declare the Orchestrator (`main`) function

5.1. `LT_DECLARE_ORCHESTRATOR(funcname) [;]`

A semicolon is not allowed if the macro is followed by its definition `{...}` of the orchestrator function; otherwise, it is required and indicates this is a declaration and a forward-reference to the orchestrator function.

funcname: - Token specifying `main`.

Termination Exit Codes: - `LT_DECLARE_ORCHESTRATOR_ERROR`

5.2. `LT_INIT_ORCHESTRATOR(funcname, id, project, size_t maxparallel);`

funcname: - Token specifying `main`.

id: - A pointer to a string specifying only alphanumeric characters which is used to identify the orchestrator function. - Default is `"o"` *if* `id` is NULL pointer or its string is empty. - A short `id` (not exceeding 3 characters) is recommended. - The `id` must be unique for the orchestrator and test group functions; a violation causes the test runner to terminate.

project: - A token identifying the project. - The token is used to generate a default report title if one is not set by the `LT_PARSE_ARGS(...)` macro.

maxparallel: Maximum number of `LT_GROUP` and `LT_TEST` macros that are allowed to execute in parallel.

Termination Exit Codes: - `LT_INIT_ORCHESTRATOR_ERROR`

5.3. `LT_PARSE_ARGS(size_t maxargs, char *customflags, char *defaultreportfilename)`

maxargs: - The maximum number arguments in the command line. - The value must 2 or greater. - The first 2 arguments and LiteTest flags are parsed by the macro. - Additional arguments must be parsed by customization code.

customflags: - A string of custom flags (e.g., `"-l-file-Q"`). - The macro causes the test runner to terminate if a non-LiteTest flag is not in the string. - The actual parsing of these must be customization code.

defaultreportfilename: - A string defining a default report file name without an extension. The extension is `.md` since the report uses Markdown Document formatting. - The maximum length including a null terminator character is `MAX_FILENAME_LEN` (129),

Termination Exit Codes: - `LT_ARG_ERROR` - `LT_FLAG_ERROR`

See “Test Command Line” for details on command-line argument and flags.

5.4. `LT_OPEN_REPORT(char *title);`

title: - A string defining the report title. - If `title` is NULL or empty, a generated default report title “Test Report” is used. - The maximum length including a null terminator character is `MAX_TITLE_LEN` (129),

Termination Exit Codes: - LT_OPEN_ERROR

5.5. LT_WRITE_RESULT(gtm, char *category);

gtm: - A LT_GROUP(...) or LT_TEST(,,) macro.

category: - A pointer to a string with the name or description of the test category. - The maximum length including a null terminator character is MAX_CATEOGRY_LEN (129).

Termination Exit Codes: LT_WRITE_ERROR.

5.6. LT_CLOSE_REPORT(char *notes);

notes: - A string of note lines to append to the test report. - Each note line must end with '\n' (newline). - If **notes** is NULL or empty, there are no notes to append. - The maximum length including a null terminator character is MAX_NOTE_LEN (10001).

Termination Exit Codes: - LT_CLOSE_ERROR

The macro writes the totals, pre-defined explanatory notes, **notes**, notes from the temporary notes file, and final summary line to the test report file, and then closes the test report.

5.7. LT_EXIT;

Exit codes: - LT_PASS (0) - LT_FAIL (1) - LT_FAULT (2) - LT_FAIL_FAULT (3)

Termination Exit Codes: - LT_EXIT_ERROR

6. Macros for a Test Group Function

Macros used to define or declare a test group function.

6.1. **LT_DECLARE_GROUP(funcname)**

funcname: - Token specifying a function name other than **main**.

A semicolon is not allowed if the macro is followed by its definition `{...}` of the test group function; otherwise, it is required and indicates this is a declaration and a forward-reference to the test group function.

Termination Exit Codes: - **LT_DECLARE_GROUP_ERROR**

6.2. **LT_INIT_GROUP(funcname, id, maxparallel)**

funcname: - The same token as the specified token for **funcname** in the preceding **LT_DECLARE_GROUP** macro defining the containing test group function.

id: - A pointer to a string specifying only alphanumeric characters which is used to identify the test group function. - Default is "**funcname**" if **id** is NULL pointer or its string is empty.
- A short **id** (not exceeding 3 characters) is recommended. - **id** must be unique for the orchestrator and test group functions; a violation causes the test runner to terminate.

maxparallel: - Maximum number of **LT_GROUP** and **LT_TEST** macros that are allowed to execute in parallel.

Termination Exit Codes: - **LT_INIT_GROUP_ERROR**

6.3. **LT_RETURN**

Termination Exit Codes: - **LT_RETURN_ERROR**

7. Macros to Execute a Test Group or a Test

- `LT_GROUP(funcname, id, [include], [isolation])[;]`
- `LT_TEST(expression, id, [include], [isolation])[;]`

A semicolon is not allowed if the macro is used as an argument to the `LT_WRITE_RESULT` macro; otherwise, it is required.

id: - A pointer to a string specifying only alphanumeric characters for the test group or test.
 - The value must be unique within the containing orchestrator or test group function - A short id (not exceeding 3 characters) is recommended or use the default. - If `id` is a NULL pointer or its string is empty, default is `n`, where `n` is 1 for the first `LT_GROUP` or `LT_TEST` macro where `id` is NULL or empty, and incremented by 1 for each subsequent `LT_GROUP` and `LT_TEST` macro where `id` is a NULL pointer or its string is empty. - See the `LT_ID` macro for obtaining the `id` for the next generated id.

include: - A single character token or omit (defaults to 1): - 0 — Never execute. This is useful to disable a test (e.g., failing or faulting test for which the fix has been deferred). - 1 — Always execute (e.g., smoke tests). - 2-9 — Execute only when the `-In` flag is specified in the test command line and `n >= include`. - I — Execute only when the `-I` flag is specified in the test command line. This is useful for injecting a fail/fault for verifying behavior and report output if fails/faults were to occur. See also `LT_FAIL` and `LT_FAULT(type)` for simulating a fail or fault, respectively.

See “Test Command Line” for details on command line flags.

isolation: - Single character token or omit (defaults to 0): - 0 — Same thread. - 1 — Separate thread. - 2 — Separate process.

See “Isolation” in the LiteTest Runner Guide.

7.1. `LT_GROUP(funcname, id, [include], [isolation])[;]`

Termination Exit Codes: - `LT_GROUP_ERROR`

7.2. `LT_TEST(expression, id, [include], [isolation])[;]`

Termination Exit Codes: - `LT_TEST_ERROR`

8. Macros for Concurrent Blocks

8.1. `LT_BEGIN_CONCURRENT(blockname)`

blockname: - Token specifying the name of the block. - Token must be the same as for the subsequent matching `LT_END_CONCURRENT(blockname)`.

Termination Exit Codes: - `LT_BEGIN_GROUP_ERROR`

8.2. `LT_END_CONCURRENT(blockname);`

blockname: - Token specifying the name of the block. - Token must be the same as for the preceding matching `LT_BEGIN_CONCURRENT(blockname)`.

Termination Exit Codes: - `LT_END_GROUP_ERROR`

9. Macros for Versioning

9.1. LT_RUNNER_VERSION

Value: - LiteTest Runner's version as a literal string of type `char[n]`, where `n` is between 5 and 9 (including the terminating null character). - Format: `M.m.p`, where `M`, `m`, and `p` are one or two digits. - `M` is the major version, `m` is the minor version, and `p` is the patch version.

The major version is incremented for major additions, removal of deprecated features, or unavoidable incompatible API changes.

The minor version is incremented for backward-compatible additions or deprecating features.

The patch version is incremented for bug fixes or internal improvements.

Examples: - `LT_RUNNER_VERSION` $\rightarrow$ `"1.0.9"` - `LT_RUNNER_VERSION` $\rightarrow$ `"1.3.11"` - `LT_RUNNER_VERSION` $\rightarrow$ `"12.05.18"`

9.2. LT_VERSION_MAJOR(v)

v: pointer to a version string.

Value: - The major version in `v` cast to `int`. - Value is `LT_INVALID_VERSION ((int)-2)` if `v` is an invalid version string.

Examples: - `LT_VERSION_MAJOR("5.0.9")` $\rightarrow$ `(int)5` - `LT_VERSION_MAJOR("5.000.9")` $\rightarrow$ `LT_INVALID_VERSION`

9.3. LT_VERSION_MINOR(v)

v: pointer to a version string.

Value: - The minor version in `v` cast to `int`. - Value is `LT_INVALID_VERSION ((int)-2)` if `v` is an invalid version string.

Examples: - `LT_VERSION_MINOR("5.00.9")` $\rightarrow$ `(int)0` - `LT_VERSION_MINOR("5.1.001")` $\rightarrow$ `LT_INVALID_VERSION`

9.4. LT_VERSION_PATCH(v)

v: pointer to a version string.

Value: - The patch version in `v` cast to `int`. - Value is `LT_INVALID_VERSION ((int)-2)` if `v` is an invalid version string.

Examples: - `LT_VERSION_PATCH("5.0.12")` $\rightarrow$ `(int)12` - `LT_VERSION_PATCH("5.000.9")` $\rightarrow$ `LT_INVALID_VERSION`

9.5. LT_VERSION_NUM(v)

v: pointer to a version string.

Value: - Representation of the version `v` cast to `int`. - Value is `LT_INVALID_VERSION ((int)-2)` if `v` is an invalid version string.

Form: MMmmpp for comparisons, e.g., 10000 for version 1.0.0, 10200 for version 1.2.0, or 11212 for version 1.12.12.

Examples: - `LT_VERSION_NUM("1.0.9")` → `(int)0x010009` - `LT_VERSION_NUM("2.5.3")` → `(int)0x020503`

9.6. `LT_VERSION_HEX(v)`

v: - Pointer to a version string.

Value: - Hex representation of version `v` cast to `int`. - Value is `LT_INVALID_VERSION` `((int)-2)` *if* `v` is an invalid version string.

Hexadecimal form `0xMMmmpp` for display/debugging, e.g., `0x010000` for version 1.0.0, `0x010200` for version 1.2.0, or `0x011212` for version 1.12.12.

Examples: - `LT_VERSION_HEX("1.0.9")` → `0x010009` - `LT_VERSION_HEX("2.5.3")` → `0x020503`

9.7. `LT_VERSION_CMP(v1, v2)`

v1: - Pointer to a version string.

v2: - Pointer to a version string.

Value: - Integer (`int`) result comparing version `v1` to `v2`: - `LT_EQUAL` `((int)0)` — equal - `LT_LESS` `((int)-1)` — `v1` is less than `v2` - `LT_GREATER` `((int)1)` — `v1` is greater than `v2` - `LT_INVALID_VERSION` `((int)-2)` — invalid `v1` or invalid `v2` - Leading zeros in `M`, `m`, and `p` are ignored during comparison.

Examples: - `LT_VERSION_CMP(LT_RUNNER_VERSION, "1.10.0")` → `LT_GREATER` (*if* `LT_RUNNER_VERSION` is `"1.10.6"`) - `LT_VERSION_CMP(LT_RUNNER_VERSION, "1.10.0")` → `LT_LESS` (*if* `LT_RUNNER_VERSION` is `"1.8.0"`) - `LT_VERSION_CMP(LT_RUNNER_VERSION, "1.10.0")` → `LT_EQUAL` (*if* `LT_RUNNER_VERSION` is `"1.10.0"`) - `LT_VERSION_CMP("1.10.0x", "1.10.0")` → `LT_INVALID_VERSION` - `LT_VERSION_CMP("1.10.0", "1.10y.0")` → `LT_INVALID_VERSION`

10. Macros for Customization

10.1. `LT_PRINT_ERR_HELP(char *err, char help)`

Print err and help text to stdout. - `err` is prefixed with `lt_err_prefix()`. - Help text is formed using `lt_usage()` and `lt_help()`.

err - Pointer to error string.

help - 0: don't print help text. - non-zero: print help text.

Note: Use `lt_set_err_prefix()` to set the prefix.

Note: Use `lt_set_usage()` and/or `lt_set_help()` to define the help text.

11. Functions for Customization

11.1. int lt_funcname(char *funcname, size_t outlen)

11.2. int lt_print_note(const char *notes)

Opens a temporary file if not already open and writes the notes to the temporary file.

notes: - A string of note lines to write to the temporary file. - Each note line must end with '\n' (newline). - If **notes** is NULL or empty, the temporary file is removed if it exists. - The maximum length including a null terminator character is MAX_NOTE_LEN (10001).

Return: - LT_SUCCESS - LT_NOTES_REMOVED - LT_NOTES_OPEN_ERROR - LT_PRINT_NOTE_ERROR

void lt_current_time(char *current_time, size_t size)

Get the current time as a formatted string.

current_time: - Buffer to store the formatted time string.

size Size of the buffer.

Note: On error, current_time is set to “unknown time”.

Customization: Get and set.

```
char *lt_executable_name(void) void lt_set_executable_name(char *en) char
*lt_default_dirpath(void) void lt_set_default_dirpath(char *dp) char *lt_err_prefix(void)
void lt_set_err_prefix(char *pe) char *lt_args_options(void) void lt_set_args_options(char
*ao) char *lt_usage(void) void lt_set_usage(char *u) char *lt_help(void) void
lt_set_help(char *h)
```

Customization Helper Functions

```
size_t lt_currentlevel(void) lt_result_t lt_currentresult(void) lt_total_t
lt_currenttotal(void) size_t lt_maxparallel(size_t level) size_t lt_currentparallel(void)
int lt_isisolated(void) int lt_isthreadisolated(void) int lt_isprocessisolated(void)
size_t lt_groupid(void) char *lt_groupname(void)
```

```
char *lt_project(void) size_t lt_maxargs(void) char *lt_title(void) size_t
lt_categoryid(void) char *lt_category(void) char *lt_funcname(void) char
*lt_notes(void) char *lt_assertexpression(void) char lt_inject(void) char
lt_isolation(void) char lt_orchestrator(void) char lt_testfunction(void) char
lt_assert(void)
```

```
char *lt_dirpath(void) char *lt_filepath(void) char *lt_filename(void)‘
```

Return: 1 true, 0 false int lt_isfilename(char *name)

Return: 1 true, 0 false, -101 invalid directory path. int lt_isreaddirpath(char *path)
int lt_iswritedirpath(char *path)

Return 1 true, 0 false, -103 invalid file path. int lt_isreadfilepath(char *path) int
lt_iswritefilepath(char *path)

12. Test Command Line

TODO.

13. Output and Golden Files

Output files in `tests/output/` and golden files in `tests/golden/` subdirectories of the repository root provide a directory layout for capturing and validating test output.

13.1. Output Files

A copy of a named output file is captured to `tests/output/` : - If the output file name has the form `.`, the `tests/output/` file is named `-.ext`. - If the output file name has the form (with no extension), the `tests/output/` file is named `-`.

A copy of `stdout` is captured in `tests/output/` output file named `stdout-<uid>` and a copy of `stderr` is captured as an output file named `stderr-<uid>`.

`<uid>` is a unique id string for each captured output file within a test run. It is used to avoid filename collisions when the same output file name appears multiple times (e.g., tests of a command line varying options each generationg `stdout` and `stderrj`).

Temporary files are not captured in `tests/output/`.

A metadata file (in Markdown Document format) for each output file is also written to `tests/output/`: - If the output file name has the form `.`, the `tests/output/` metadata file is named `-.ext.md`. - If the output file name has the form (with no extension), the `tests/output/` file is named `-.md`.

An output file and its metadata file in `tests/output/` may be copied (promoted) to the `tests/golden/` subdirectory of the repository root directory. See the following section.

13.2. Golden Files

A metadata golden file having a name with the form `.ext` applies to all golden files named `-ext` and can be added manually or by customization code added to the test runner.

A metadata golden file having a name with the form `.md` applies to all golden files named `-` with no extension and can be added manually or by customization code added to the test runner.

Golden files (located in the `tests/golden/`) are handled as follows:

- If the corresponding golden file with the same name does not exist,
 - If a metadata golden file `.ext` or exists and contains the line `##MATCH##`, comparison is skipped and a match is forced. If it contains the line `##MISMATCH##`, comparison is skipped and a mismatch is forced.
 - Otherwise, the output file is automatically copied (promoted) into the `tests/golden` directory and a match forced.
 - The test report notes that the file was promoted.
- If the corresponding golden file exists and contains the line `##MATCH##`, comparison is skipped and a match is forced.
- If the corresponding golden file exists and contains the line `##MISMATCH##`, comparison is skipped and a mismatch is forced.
- Otherwise, the output file is compared with its existing golden file of the same name in the `tests/golden` directory to determine whether they match.

If an output file does not match its golden file but inspection determines the output is correct, update the golden file with a copy of the output. Alternatively, remove the golden file so that the next run will automatically promote the output.

13.3. Output/Golden Rationale

LiteTest uses a parallel `tests/golden` directory with golden files that keep the exact same filename as their corresponding test output. This design avoids the common problems found in extension-based naming schemes:

- **No filename collisions** Different outputs such as `foo.c` and `foo.h` remain distinct (`foo-<uid>.c` vs. `foo-<uid>.h`) instead of collapsing into a single `foo.golden`.
- **No special-case rules** The golden filename is always identical to the output filename.
- **Preserves file type information** Because the original extension is retained, it is immediately clear whether a golden file contains text, Markdown Document, JSON, HTML, binary data, or something else.
- **Simpler mental model** Developers do not need to remember naming conventions or extension-replacement rules. Golden files simply live in a parallel directory and share the same name as the output they validate.